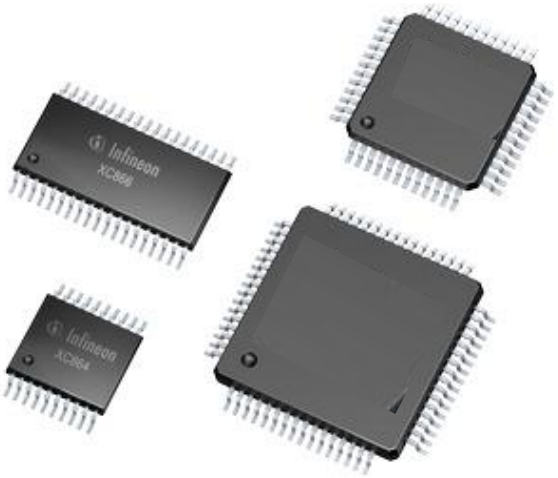
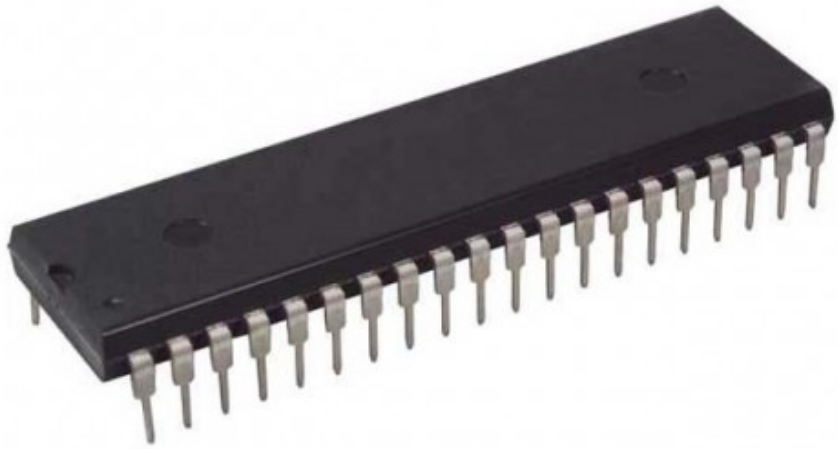
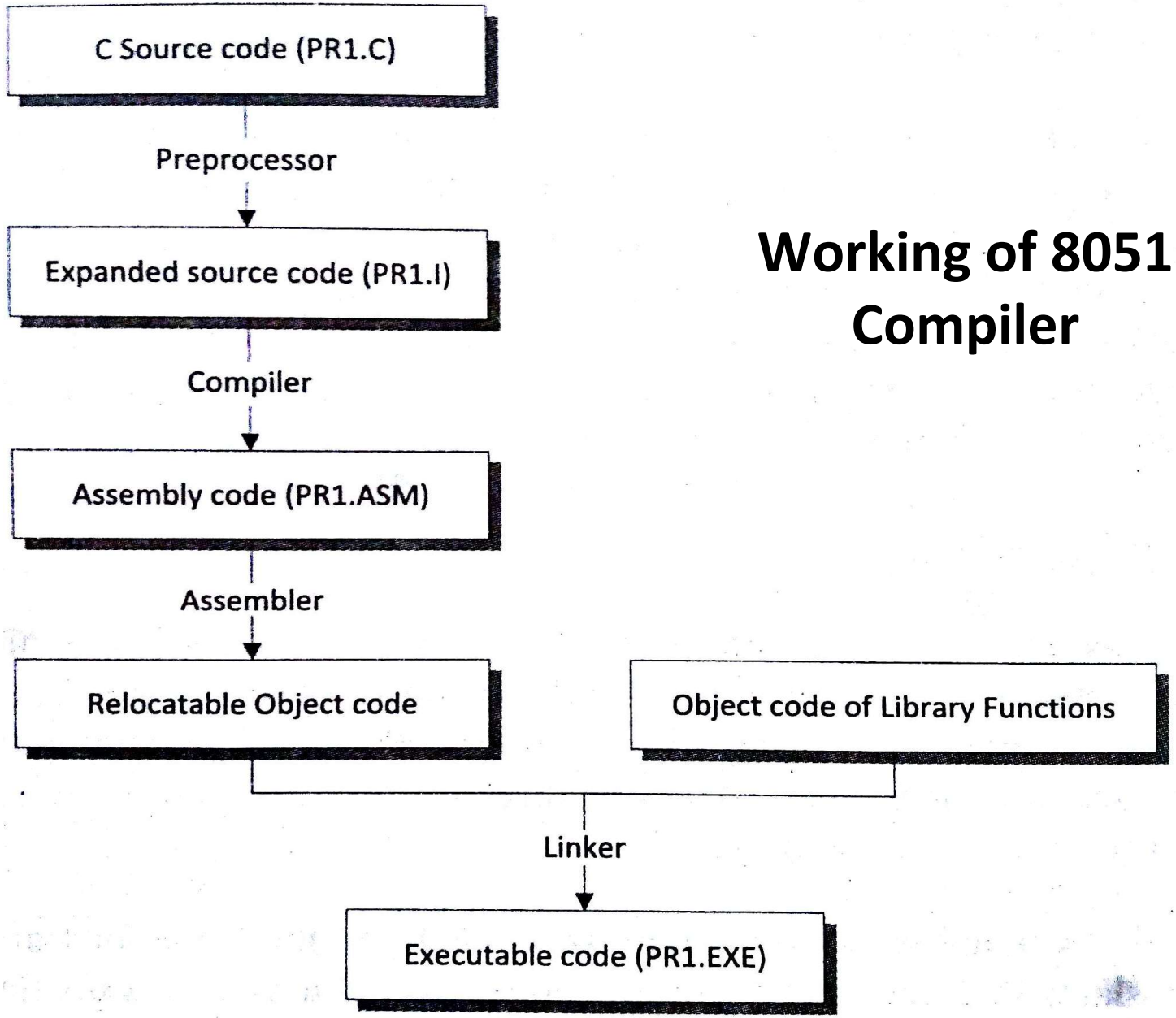
8051 Microcontroller



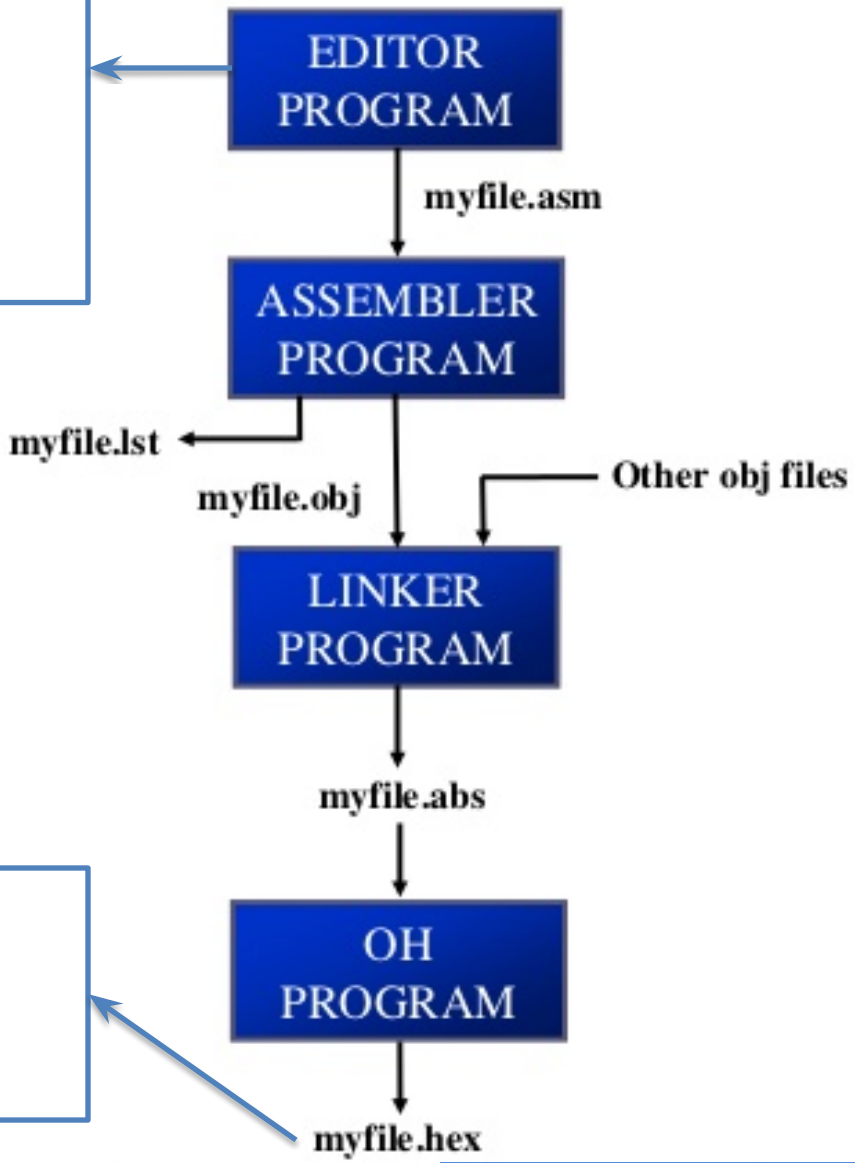


Working of 8051 Compiler



```
HERE:    MOV    R0,#01H
         MOV    R1,#02H
         MOV    A,R0
         ADD    A,R1
         MOV    P0,A
         SJMP   HERE
```

Working of 8051 Assembler




```
:03000000020800F3
:0C080000787FE4F6D8FD75810702000047
:0A00000078017902E829F58080F606
:00000001FF
```

```
#include<reg51.h>
int main()
{
while(1)
{
char R0=1,R1=2,A;
P0=R0+R1;
}
return 0;
}
```

```
HERE:    MOV    R0,#01H
         MOV    R1,#02H
         MOV    A,R0
         ADD    A,R1
         MOV    P0,A
         SJMP   HERE
```

Address	Opcode	Operand
0000	78	01
0002	79	02
0004	E8	
0005	29	
0006	F5	80
0008	80	00

Note: Address of P0 = 80h